# CHAPTER 6

## Kramers–Kronig Relations

**General relation between the real and imaginary parts of $\tilde{n}$ or $\varepsilon$**



- Now just go to time and real space .
- The most general expression for a linear response function is :

$$\frac{1}{\varepsilon_0}\boldsymbol{P}(\boldsymbol{r},t)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\chi(\boldsymbol{r},\boldsymbol{r}',t,t')\boldsymbol{E}(\boldsymbol{r}',t')dt'd\boldsymbol{r}' \qquad (6.1)$$

The polarization ***P*** at point ***r*** and time *t* depends on the electric field at all other places and at all times

First, we assume the sample is homogeneous in time, $\chi$ depends only on $t-t'$ ;

Second, $\boldsymbol{E}(\boldsymbol{r}',t')$ are much longer than the lattice constant, depends only on $\boldsymbol{r}-\boldsymbol{r}'$ :

$$\frac{1}{\varepsilon_0}\boldsymbol{P}(\boldsymbol{r},t)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\chi(\boldsymbol{r}-\boldsymbol{r}',t-t')\boldsymbol{E}(\boldsymbol{r}',t')dt'd\boldsymbol{r}' \qquad (6.2)$$

$$\varepsilon_1(\omega)-1=\mathrm{Re}\{\chi(\omega)\}=\frac{2}{\pi}P\int_0^{+\infty}\frac{\omega'\varepsilon_2(\omega')}{\omega'^2-\omega^2}d\omega'$$

$$\varepsilon_2(\omega)=\mathrm{Im}\{\chi(\omega)\}=-\frac{2\omega}{\pi}P\int_{-\infty}^{+\infty}\frac{\varepsilon_1(\omega')-1}{\omega'^2-\omega^2}d\omega'$$



$$n(\omega)-1=\frac{2}{\pi}P\int_0^{+\infty}\frac{\omega'\kappa(\omega')}{\omega'^2-\omega^2}d\omega'$$

如果无吸收
$n=1$

$$\kappa(\omega)=-\frac{2\omega}{\pi}P\int_{-\infty}^{+\infty}\frac{n(\omega')-1}{\omega'^2-\omega^2}d\omega'$$

**Kramers–Kronig relations**

- If $\varepsilon_1(\omega)$ or $n(\omega)$ deviate in some frequency range from 1, then there must necessarily be absorption structures somewhere.
- if either the real or imaginary part of $\varepsilon(\omega)$, $\tilde{n}(\omega)$ is known over the whole spectral range, t hen the other part can be calculated.